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Indicators of Pollution in *Monteverde Streams*

JUNE 19, 2025

ADOPT-A-STREAM PROJECT

- Program started in 2009 to gather information about streams in the Monteverde area
- Data collected by every two months students or volunteers
- Streams included: *Quebrada Colegio, Q. Cuecha, Q. La Máquina, Q. Sucia, Q. La Saca, and Río Negro*



SOURCE: LUISA MORENO SCOTT,
WWW.MONTEVERDE-INSTITUTE.ORG

Research Questions:

1. Is there a relationship between the abundance of some taxonomical families within good quality streams over time?
2. Is the abundance of some families in good quality streams affected by rainfall?

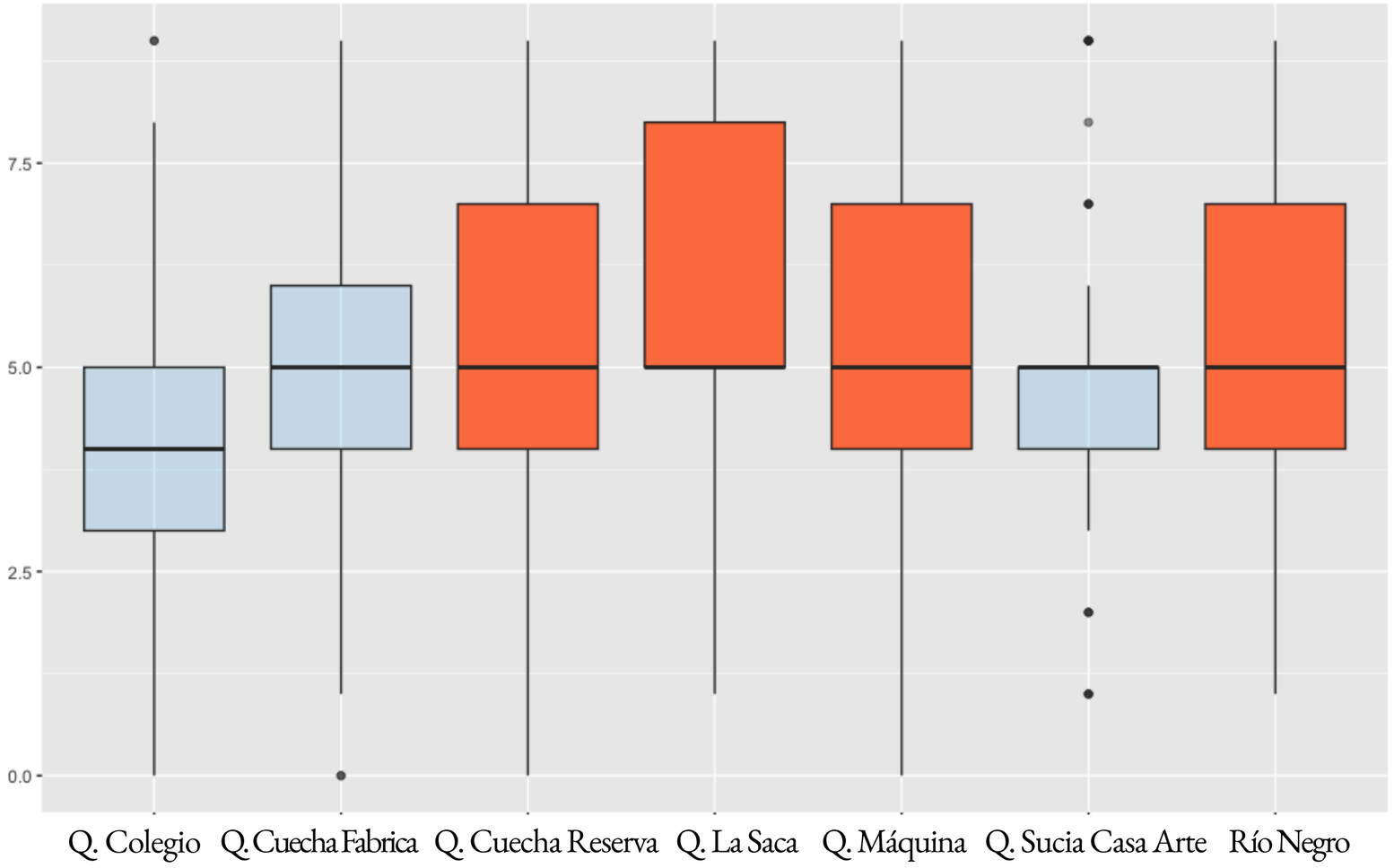
DATA

- 2,400 observations collected between 2018 and 2024
- Sample size after removing missing data: **1,389**
- Bioindicator of water quality: Biological Monitoring Working Party (**BMWP**) Index of species found in the water
  - Higher value means higher tolerance to pollution
- Measured quantities: water temperature, dissolved oxygen, conductivity, pH, flow

DATA

Streams and taxonomic families of interest:

ORDER	FAMILY
Thricoptera	Hidrobiosidae
	Lepidostomatidae
	Calamoceratidae
	Leptoceridae
	Philopotamidae
Coleoptera	Ptilodactylidae
	Elmidae
Odonata	Polythoridae
Ephemeroptera	Heptageniidae
Crustacea	Hyallelidae



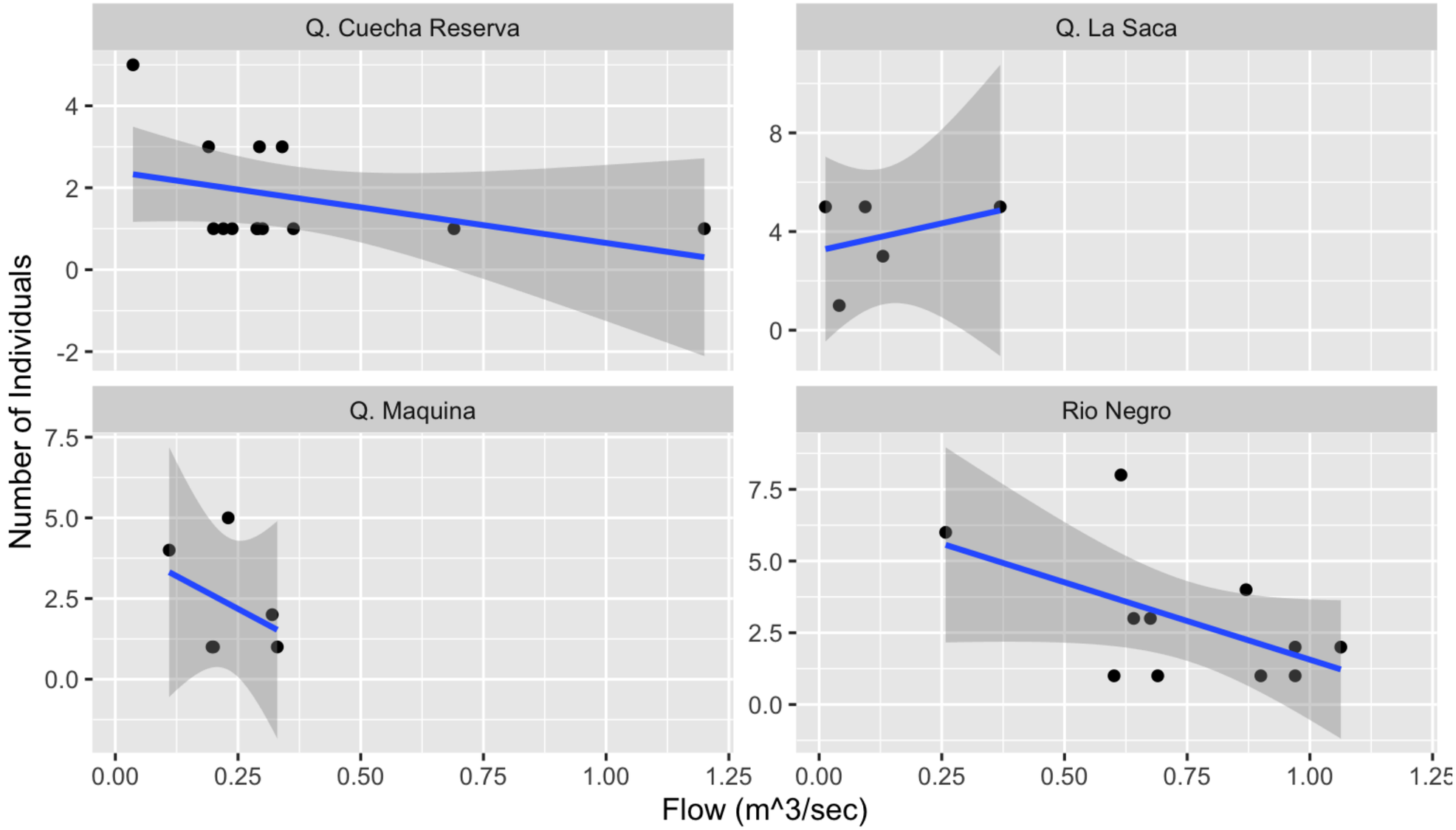
ANALYSIS

- *Response variable:* observed number of individuals
- *Predictor variables:* date of observation, stream flow rate, monthly precipitation in Monteverde
- Linear regressions to test for relationships between response and predictor

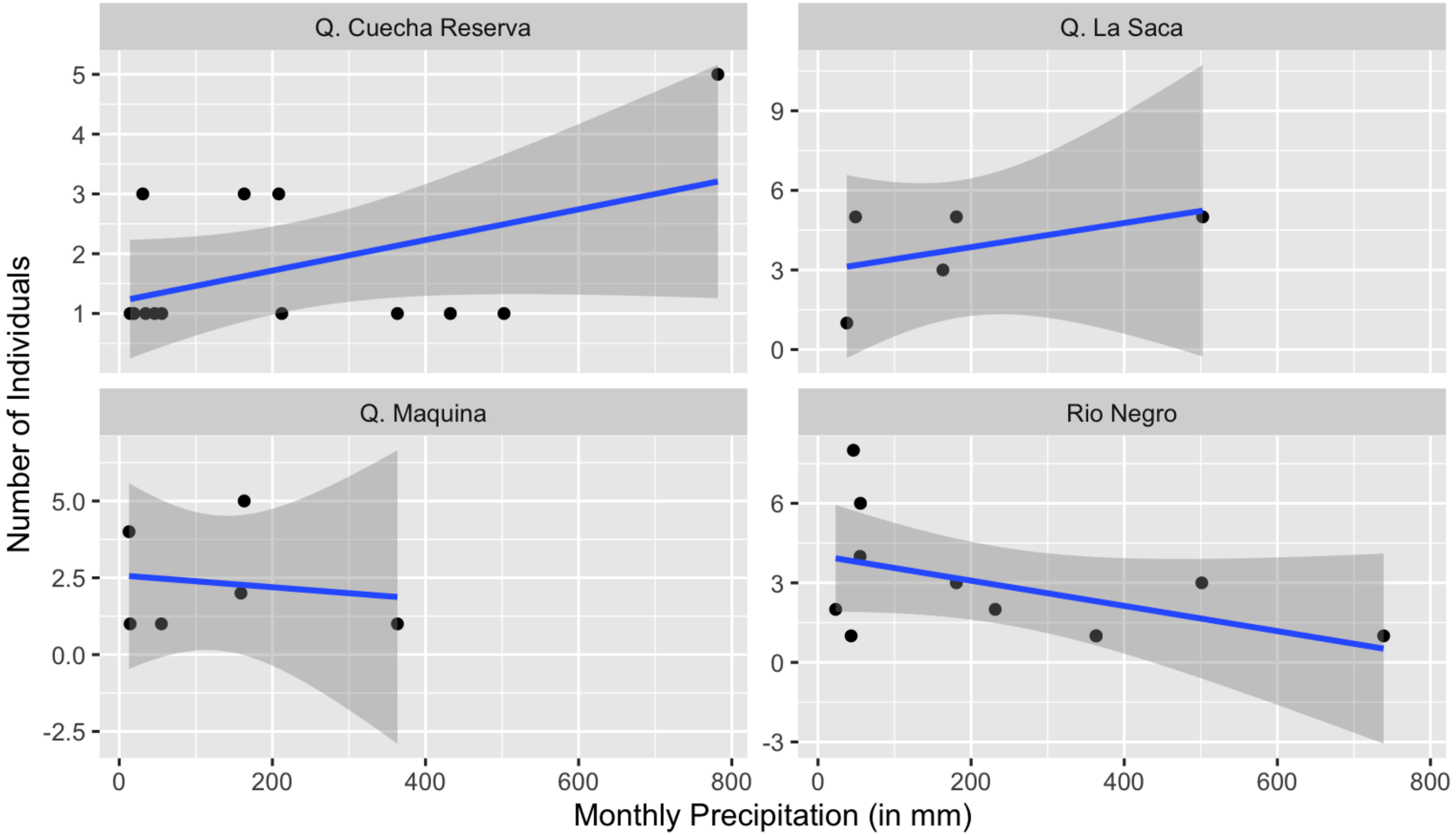
PROPORTION OF DETECTED ASSOCIATIONS:

Variable	Detection proportion
Monthly precipitation:	15%
Flow:	10%
Date:	0%

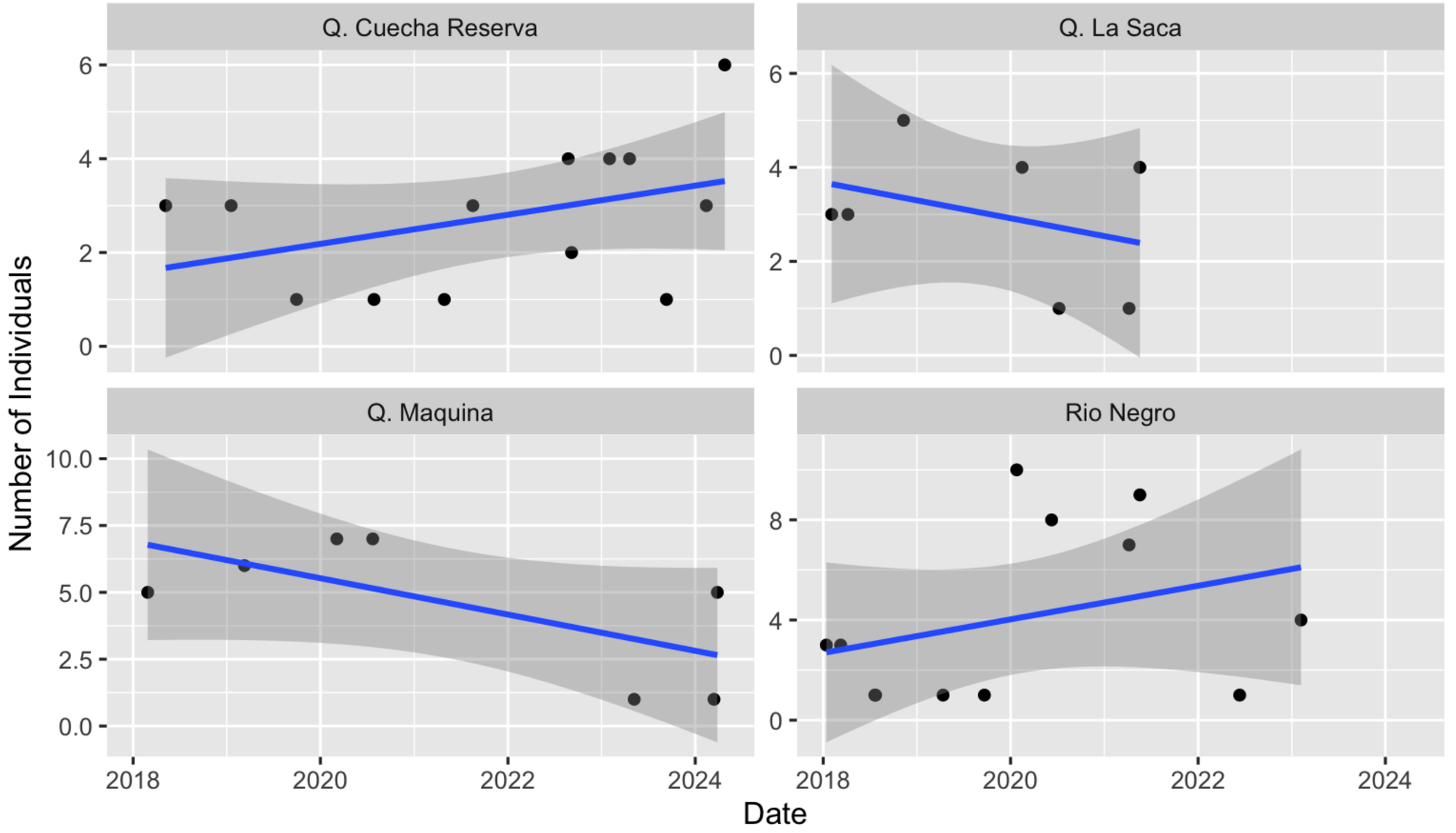
Coleoptera Elmidae and Rainfall (flow)



Coleoptera Elmidae and Rainfall (monthly precip. in Monteverde)



Apperance of Coleoptera Elmidae over time



Conclusion

There is little evidence that date or rainfall are associated with family abundance.

- Most cases were not statistically significant although visual trends were observed
- Possible limitations:
  - Sample size
  - Potential issues with correlation between predictor variables

Questions?

Thank you!